

SEDAR 64 SAFMC & GMFMC Yellowtail Snapper

DATA WORKSHOP RESEARCH RECOMMENDATIONS

LIFE HISTORY RESEARCH RECOMMENDATIONS

1.1.1. Stock Definition

- Investigate the genetic linkages of Yellowtail Snapper populations between Florida and the Carolinas and between the Gulf of Mexico and western Caribbean.
- Investigate the current occurrence of hybrids (e.g., with Lane Snapper) throughout the range of the stock.

1.1.2. Natural Mortality

- As the apparent maximum age of Yellowtail Snapper increased from assessment to assessment, the natural mortality estimates decreased. Estimates of natural mortality that are derived independently from life history parameters would help to validate these methods. Given adequate fishery independent age information, total mortality (fishing mortality plus natural mortality) can be estimated. In addition, telemetry and tag-recapture methods can offer independent estimation of fishing mortality and natural mortality, however these methods rely on high site fidelity of Yellowtail Snapper to reef sites or reliable tag return rates.
- Investigate estimates of natural mortality rates for different life stages of Yellowtail Snapper using ecosystem simulation models (e.g., Ecopath with Ecosim and OSMOSE).

1.1.3. Release Mortality

- On-board observers inform immediate release mortality, however information on delayed mortality is limited. Additional tagging of Yellowtail Snapper with passive and acoustic tags, as well as the continued development of tag-and-recapture models would help to inform delayed release mortality.

1.1.4. Age and Growth

- Expand and increase the amount of length-at-age data coming from fishery-independent biological sampling throughout the range of the stock (especially for fish smaller than the current minimum size limit).
- Continue to sample the population off the Carolinas undergoing reduced targeted fishing pressures and allowing for greater estimates of maximum age.

1.1.5. Reproduction

- Expand information on reproductive characteristics such as age- and size-at-maturity, fecundity, sex ratio, and distribution of spawning aggregations throughout the range of the stock.

1.1.6. Movements and Migrations

- Investigate juvenile ontogenetic shifting from nearshore areas to reef habitat.
- Investigate movement and migration rates between the Florida Keys, southeast Florida, and southwest Florida (e.g. acoustic tagging and stable isotope studies).

COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

Improve or develop new methods for collecting discard data. Expand observer coverage to the entire range for Yellowtail Snapper (i.e. Atlantic) to document discard length and mortality.

Find a better method to address false zeros in self-reported logbook data. Explore recall bias/rounding issue: discards 5,10, 15 – recall bias – 1-10, units of 5 after that.

Study smaller fish for possible correlation between sex and tail length. Industry has seen robust fish with short tails and skinny fish with longer tails and believe them to be evidence of a secondary sex characteristic.

Perform genetic analysis of commercial samples to determine if Yellowtail Snapper is a single stock in the Southeastern United States (very old and large fish North of Florida along the Atlantic coast possibly indicating different stocks).

So little data is available on YOY/juvenile Yellowtail Snapper. There may be an opportunity to increase these samples as commercial fishers who participated in the workgroup have offered to assist fisheries scientists to obtain samples of YOY/juvenile Yellowtail Snapper. Industry believes they can get fisheries independent scientists' access to these fish by taking scientists to areas where many YOY/juvenile fish have been observed, or by providing them with area and gear recommendations based on the results of commercial fishing activities for Yellowtail Snapper.

Survey fishers for when they encounter small sub-legal fish (on board observer or email/mail). When they see small fish, they often leave the site which is not captured by logbook or gulf observer program. Modifying API of e-logbook or putting more onboard observers in the keys could provide more data on behavior. Onboard observers could also obtain discard information. Could use VMS to account for target species switching.

Ensure consistent and adequate levels of funding for continued TIPS sampling. These data were critical in providing age, length, weight, and trip information which can help validate reported landings information.

RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- Continue to collect discard length and age data from headboat and charterboat sectors.
- Increase research efforts to collect discard and retained length and age data from the private sector.
- Increase at-sea observer coverage for nighttime trips.
- Assess the impact of headboats that do not renew their federal reef fish permits and target popular reef fish species solely in state waters on the SRHS coverage.
- Recommend methods to estimate uncertainty in headboat landings and discards.

MEASURES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS During the review and evaluation of the various program datasets and indices presented during the Data Workshop, the PAW identified the following research recommendations to further improve the indices of relative abundance:

- Develop fishery-independent surveys throughout the Florida Keys which successfully target settlement sized Yellowtail Snapper in seagrass/mangroves habitats before ontogenetically shifting to reef habitats. This habitat shift is observed throughout the Caribbean but not well documented for Florida.
- Develop or extend fishery-independent reef fish surveys into deeper waters (>30 m) along the Florida Keys for greater overlap with exploited portions of the population.

ASSESSMENT WORKSHOP RESEARCH RECOMMENDATIONS

Age and Growth

Age data for southeastern U.S Yellowtail Snapper suggest there may be multiple growth patterns, such that fish beyond the Florida Keys may grow faster and have larger asymptotic sizes. Data are sparse however outside of South Florida, limiting these speculations. Therefore, a recommendation is to increase otolith sampling outside of South Florida and to explore alternative model configurations that allow for multiple growth patterns (e.g. multiple areas, areas-as-fleets). It is also recommended to increase otolith sampling for private and charterboat modes which are highly under represented. Lastly, it is recommended to explore methods to weight the age data sampled from landings accordingly to account for regional differences and uneven sampling of the landings.

Length Composition

For length samples to be a better representation of the length composition of landings and discards, it is recommended to increase sampling of lengths in regions outside of the Florida Keys as presented in S64-AW-01. Length compositions of discards are valuable model inputs, therefore it is recommended to continue data collection from at-sea observer programs and to expand the coverage of these programs. This was also

a research recommendation discussed in SEDAR 27A. Additionally, it is recommended to increase length sampling for private and charter recreational modes which are highly under represented.

Commercial Discards

Commercial discards are currently highly uncertain. It is recommended to explore data collection and data analysis methods to increase precision on these estimates.

Headboat Landings and Discards

Uncertainty of headboat landings and discards are unknown and should be evaluated. Additionally, some headboats in South Florida are exempt from participating in the SRHS and no federally administered surveys have absorbed these vessels into their sample frames, eliminating opportunities for these vessels to report landings or fishing effort.

Fishery Independent Data

Age samples for Yellowtail Snapper from fishery independent sources are lacking and would be highly useful in determining growth. In addition, an index of abundance of young-of-the-year (<10 cm FL) Yellowtail Snapper targeted in seagrass habitat would aid in refining the recruitment signal over time of this species.

REVIEW PANEL RESEARCH RECOMMENDATIONS

ToR #6: Consider the research recommendations provided by the Data and Assessment workshops and make any additional recommendations or prioritizations warranted.

- ***Clearly denote research and monitoring that could improve the reliability of, and information provided by, future assessments***

More analysis and synthesizing of existing biological information pertaining to spatial stock structure and dynamics is needed. The stock occurs at the edge of the species' range and cursory

analysis suggests that movement, migration, and life history plasticity are important factors in stock dynamics. Additionally, further investigation of stock structure, spawning areas, larval transport and juvenile/adult movement using methods such as otolith microchemistry or stable isotopes would be useful.

Age validation studies are needed to test whether growth checks are laid down consistently throughout the area of distribution and sampling (which includes tropical and temperate habitats) and reflect annual increments.

Age-length sampling among areas of the stock distribution is needed. Altered age-length sampling may require re-allocating sampling effort from the FL Keys and Southeast FL to other areas. These samples would improve growth information representing data throughout the range of the stock. This may be informed by outcomes of the analyses suggested above.

Improve sampling of discards, particularly in the commercial and headboat sectors.

Consider options for improving fisheries independent sampling of the yellowtail snapper stock.

- ***Provide recommendations on possible ways to improve the SEDAR process***

There is a need to consistently outline reasoning behind decisions made in the assessment process and reference relevant information sources such as data workshop reports.

Provide presentation files (PowerPoints) in advance of workshop sessions and include file name and page number on every slide to facilitate referencing of slides in discussions. This can also include past assessment reports

(https://www.nefsc.noaa.gov/saw/reviews_report_options.php).